# TIM-MGDT Phase 10 - GPU Execution Instructions

This guide outlines the precise steps, settings, and datasets used to run the Phase 10 evaluations for **Mistral-7B** and **Llama-3-8B** on your 2x RTX 3090 GPU remote setup.

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## 📋 Exact Execution Settings & Datasets

To ensure maximum speed while providing robust, publication-grade results, both scripts are fully optimized with the following configurations:

- **Dataset Size: Large rigor suite** with **500 samples** for inference benchmarking and **100 samples** for deep layer-importance attention analysis.
- **Batch Size:** Batched to **16** (instead of sequential batch size = 1). This utilizes the full parallel computing power of your 3090s, offering a **10x–20x speedup** while keeping VRAM well within safety limits.
- **Precision:** Loaded in **16-bit Float (`fp16`)** utilizing PyTorch native scaling.
- **Device Layout:** Pinning models to separate cards (`CUDA_VISIBLE_DEVICES`) to enable independent parallel execution.
- **Zero GPT-2 Fallbacks:** Runs only the target model (`meta-llama/Meta-Llama-3-8B` and `mistralai/Mistral-7B-v0.1`). If the gated model requires validation stubs, it fails cleanly with a clear status instead of sliding back to GPT-2.

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## 🛠️ Execution Scripts (Virtual Environment Autoloading)

We have created dedicated scripts under the `runners/` directory. All scripts automatically detect your virtual environment (`venv` or `.venv`) at the repository root, activate it, pin the designated CUDA devices, and run the pipeline.

### Option A: Run Mistral-7B (Only on GPU 0)

Pins execution to **GPU 0** so it completes independently:

```
chmod +x runners/run_mistral.sh
./runners/run_mistral.sh
```

### Option B: Run Llama-3-8B (Only on GPU 1)

Pins execution to **GPU 1** so it completes independently:

```
chmod +x runners/run_llama.sh
./runners/run_llama.sh
```

### Option C: Run Both Simultaneously in Parallel

To utilize both 3090s simultaneously so that **one finishes early** without waiting for the other, use our parallel orchestrator:

```
chmod +x runners/run_parallel.sh
./runners/run_parallel.sh
```

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## 📊 Result Capture & Debugging for Publication

Upon completion, all metrics are stored inside the `experiments/phase10/` directory:

1. **Mistral-7B Results:** `experiments/phase10/mistral7b_real_results.json`
2. **Llama-3-8B Results:** `experiments/phase10/llama3_8b_real_results.json`

### Captured Metrics Include:

- Precise execution time and timestamps
- Throughput (**tokens/sec**) and average batch latency (**ms**)
- Peak GPU memory consumption (**GB**) and memory growth gradients
- Layer-importance profiles (sorted attention weight standard-deviations)
- 4-Stage Lottery Ticket analysis (Pruned params, quality preservation scores)
- `actual_gpu_computation` **verification flag** (proof of real hardware execution, not instant stubs, for publication review).